Generalization through differential privacy

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An elementary proof of the transfer theorem

A New Analysis of Differential Privacy's Generalization Guarantees

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Saeed Sharifi-Malvajerdi Moshe Shenfeld

September 10, 2019

Abstract

We give a new proof of the "transfer theorem" underlying adaptive data analysis: that any mechanism for answering adaptively chosen statistical queries that is differentially private and sample-accurate is also accurate out-of-sample. Our new proof is elementary and gives structural insights that we expect will be useful elsewhere. We show: 1) that differential privacy ensures that the expectation of any query on the *posterior distribution* on datasets induced by the transcript of the interaction is close to its true value on the data distribution, and 2) sample accuracy on its own ensures that any query answer produced by the mechanism is close to its posterior expectation with high probability. This second claim follows from a thought experiment in which we imagine that the dataset is resampled from the posterior distribution after the mechanism has committed to its answers. The transfer theorem then follows by summing these two bounds, and in particular, avoids the "monitor argument" used to derive high probability bounds in prior work.

An upshot of our new proof technique is that the concrete bounds we obtain are substantially better than the best previously known bounds, even though the improvements are in the constants, rather than the asymptotics (which are known to be tight). As we show, our new bounds outperform the naive "sample-splitting" baseline at dramatically smaller dataset sizes compared to the previous state of the art, bringing techniques from this literature closer to practicality.

https://arxiv.org/abs/1909.03577.

Adaptive data analysis

Given a dataset S, we wish to perform a sequence of analyses,

$$q_1, q_2, \ldots, q_k$$

where the queries q_t adapt to the previous answers.

- **Problem:** reusing data can lead to overfitting

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  - amount of data grows linearly with number of queries

Connection to differential privacy

A **differentially private mechanism** is a randomized algorithm where the distributions of the outputs computed from similar datasets are also similar.

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- Intuition: if an analysis is differentially private, then answers generalize since they don't depend closely on the particular dataset

Transfer theorem

Informal theorem. A differentially private analysis that has high *in-sample* accuracy must also have high *out-of-sample* accuracy.

Transfer theorem: prior works

Proofs

- Original connection to differential privacy: [DFHPRR15]
- Best analysis via the 'monitor argument': amount of data grows $\sqrt{k}$ with respect to number of queries k. [BNSSSU16]

Lower bounds

- The analysis in [BNSSSU16] is asymptotically tight, as seen in [HU14], [SU15]. This work [JLNRSS19] improves concrete bounds through new proof techniques.

Transfer theorem: this work

Adaptive data analysis consists of the following generating process:

1. collect a dataset S from an underlying distribution $\mathcal{P}^n$

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1. collect a dataset S from an underlying distribution $\mathcal{P}^n$
2. interact with data to produce transcript Π of interaction

Transfer theorem: this work

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1. sample an interaction Π

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1. sample an interaction Π
2. sample a dataset S' from the posterior distribution conditioned on the interaction Π

Transfer theorem: this work

Sketch of argument

1. The Bayesian resampling lemma implies that adaptive analysis is equivalent to non-adaptive analysis over posterior.

Transfer theorem: this work

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Preliminaries: setting

Ingredients

- $\mathcal{X}$ an abstract data domain

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- $S = \{S_i\}_{i=1}^n \in \mathcal{X}^n$ a collection of n data records
- Q is a family of queries q
- $q : \mathcal{X}^* \to [0, 1]$ a linear data query:

$$q(S) = \frac{1}{n} \sum_{i=1}^{n} q(S_i).$$

Preliminaries: setting

Additional notation

- Let S_i denote the random variable and x its realization

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- If $\mathcal{D}$ is a distribution over datasets, $q(\mathcal{D})$ is the expectation:

$$q(\mathcal{D}) = \mathop{\mathbb{E}}_{S \sim \mathcal{D}}[q(S)].$$

Preliminaries: adaptive analysis

Interacting parties

- $\mathcal{A} : \mathbb{R}^* \to Q^*$ an analyst

Preliminaries: adaptive analysis

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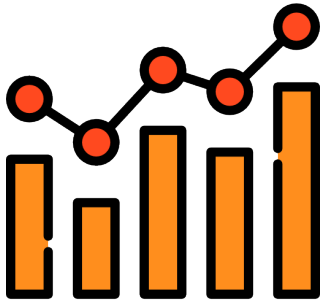
- $\mathcal{A} : \mathbb{R}^* \to Q^*$ an analyst
- $M : \mathcal{X}^n \times Q^* \to \mathbb{R}^*$ a (possibly stateful) statistical estimator

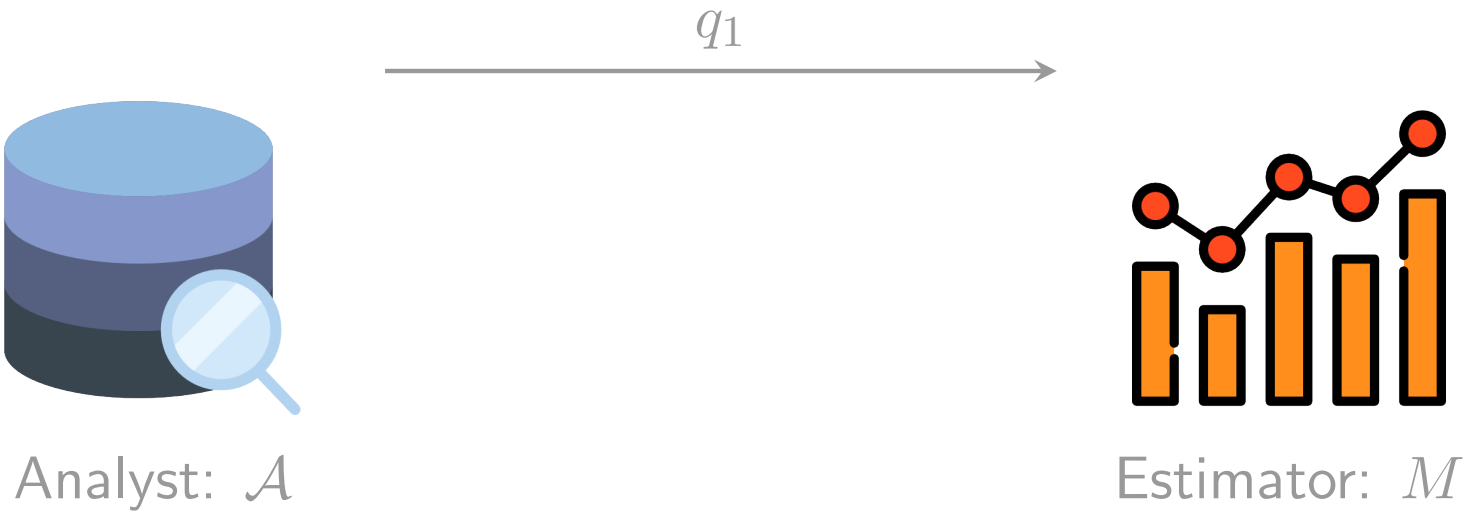
Preliminaries: adaptive analysis

Analyst: $\mathcal{A}$



Estimator: M

Preliminaries: adaptive analysis



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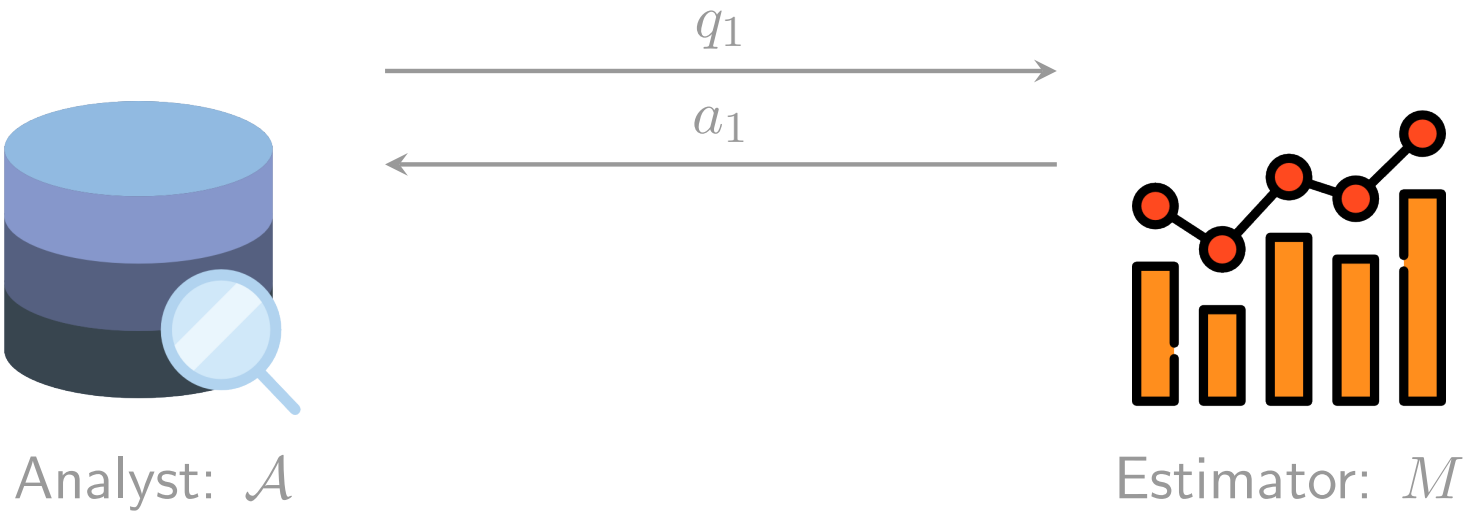

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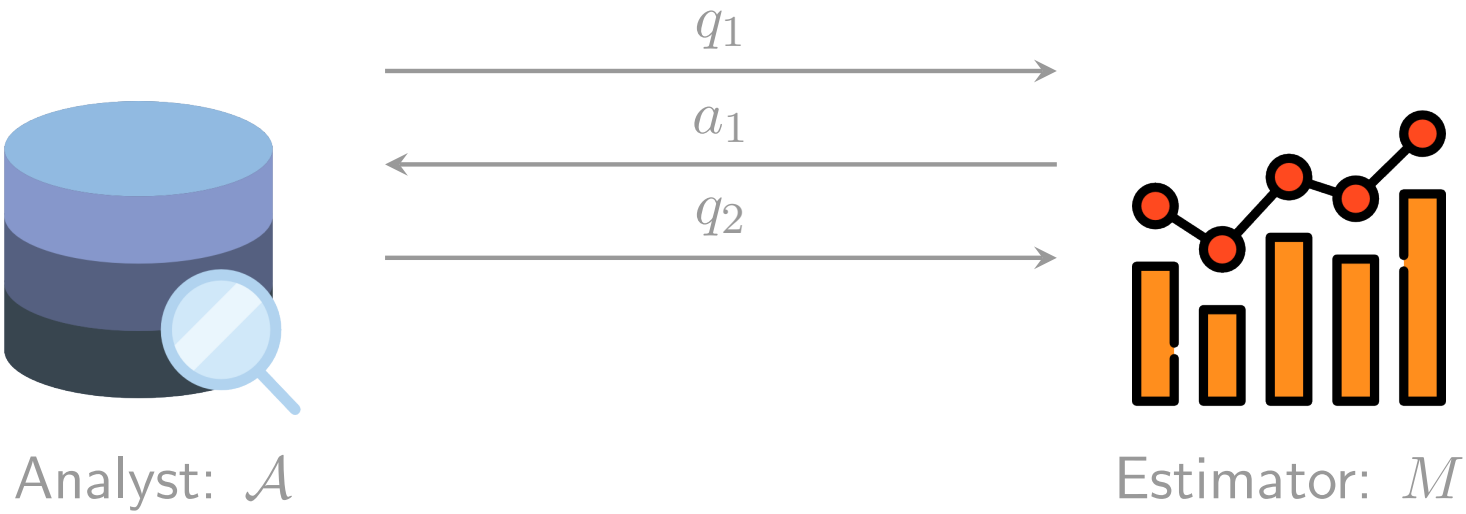

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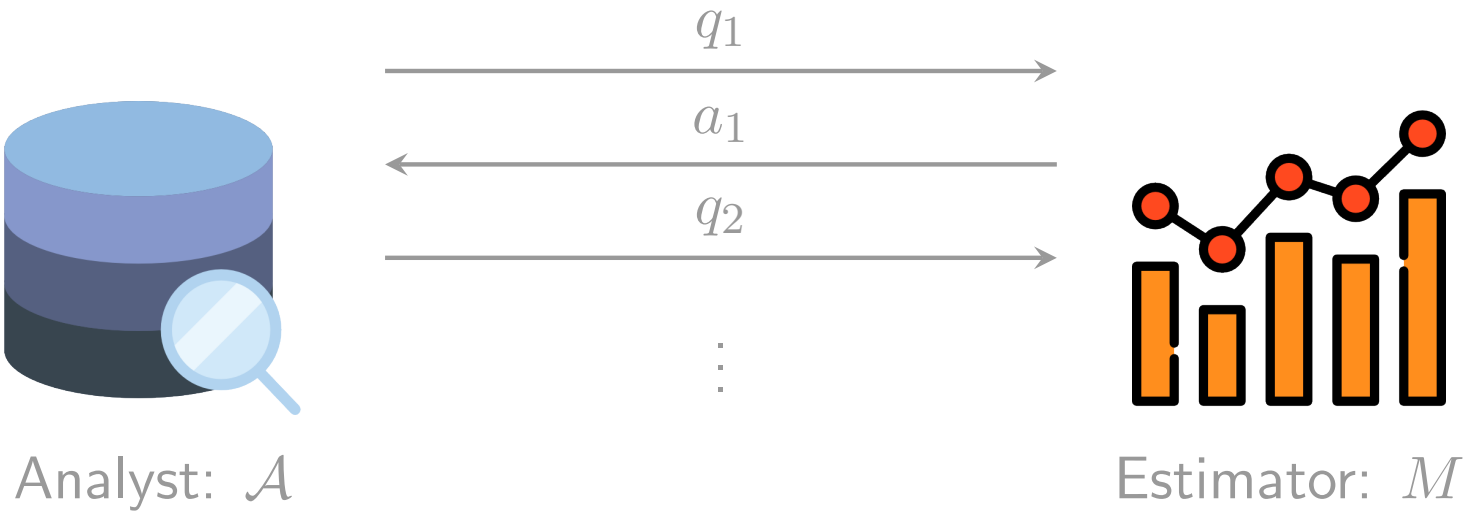

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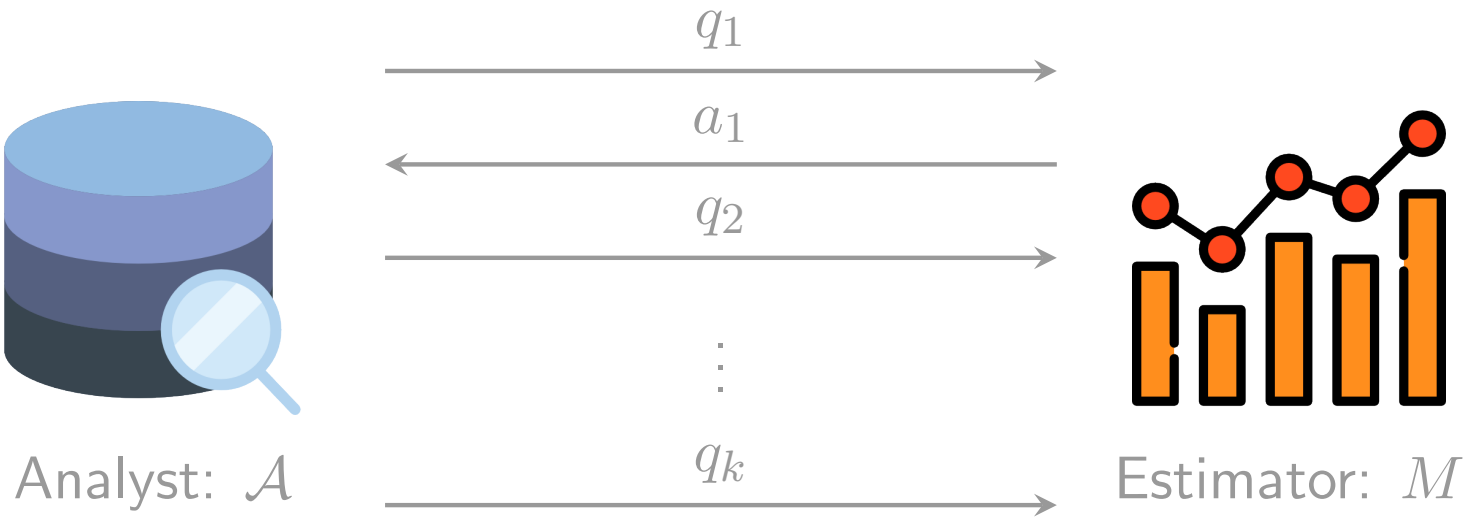

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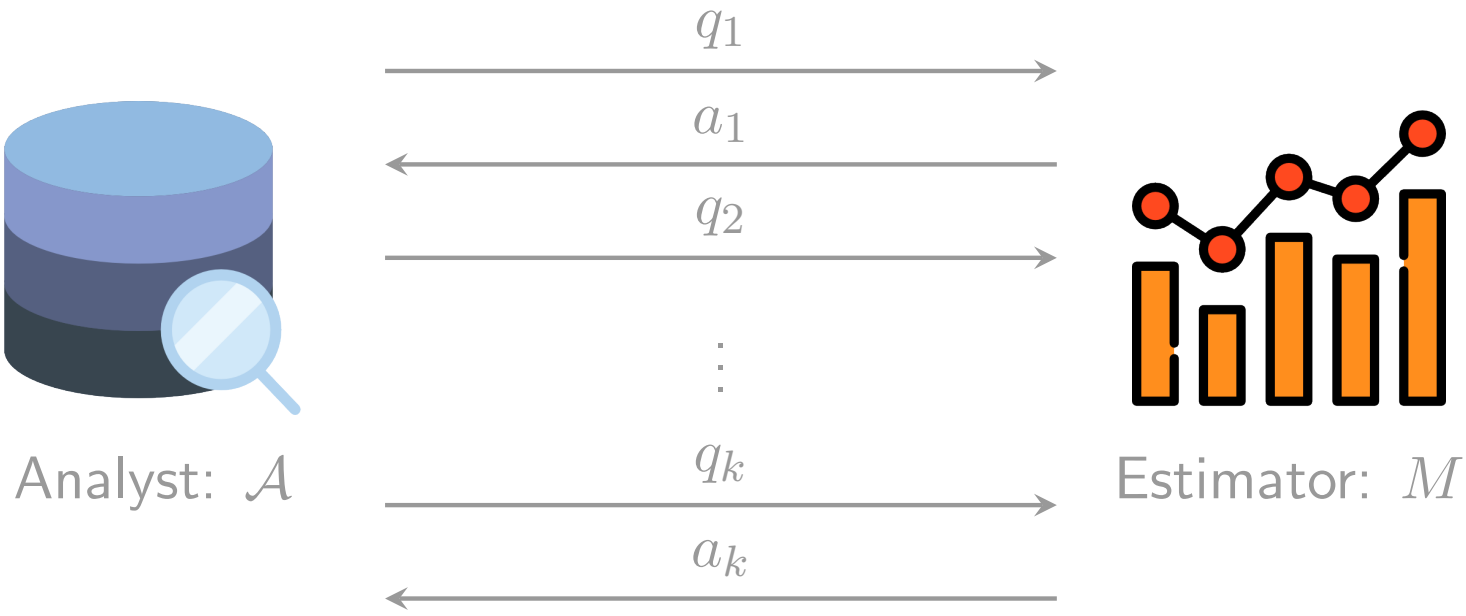

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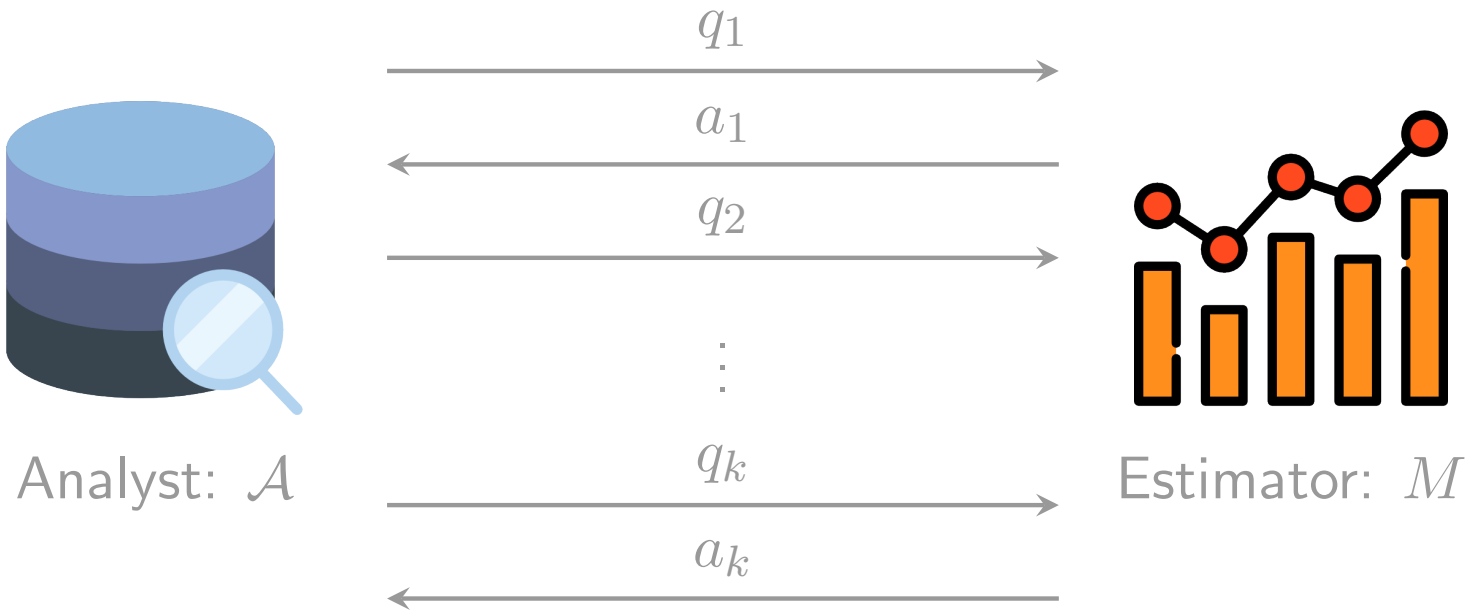

Transcript: $\pi = \{(q_1, a_1), \ldots, (q_k, a_k)\}$.

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Notation

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- Π denotes the random variable and π its realizations
- $\mathrm{Interact}(M, \mathcal{A}; S)$ is the transcript of the interaction
  - for brevity, abbreviate $\mathrm{Interact}(M, \mathcal{A}; S)$ by $I(S)$

Preliminaries: adaptive analysis

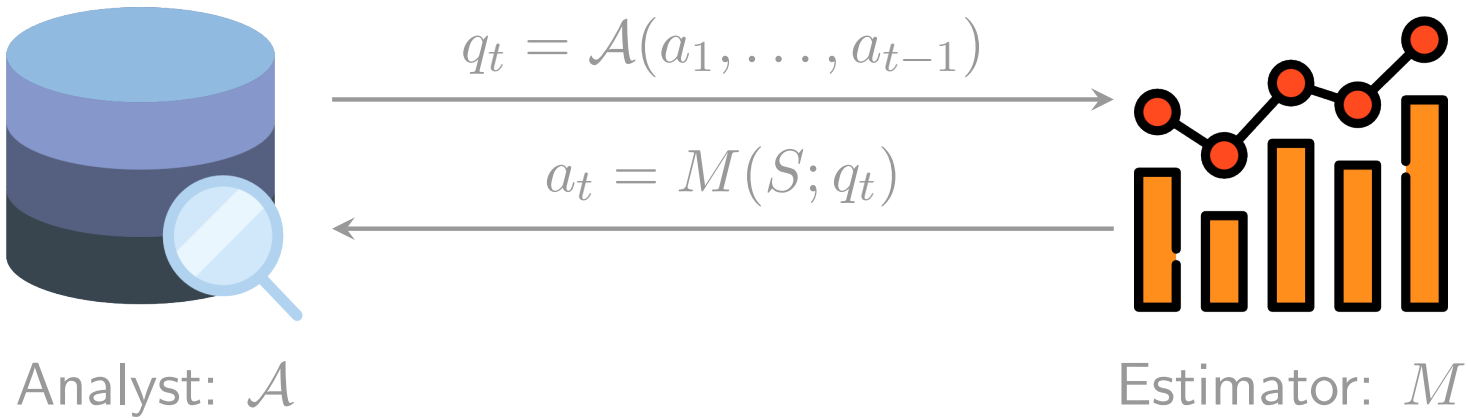

Figure 1: The interaction between $\mathcal{A}$ and M on dataset S generates a transcript $\mathrm{Interact}(M, \mathcal{A}; S) \in \mathbf{\Pi}$.

Preliminaries: product and posterior distribution

We consider datasets S drawn from two distributions, $\mathcal{P}^n$ and $\mathcal{Q}_\pi$:

- $\mathcal{P}^n$ is the product distribution over datasets with n records

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- $\mathcal{P}^n$ is the product distribution over datasets with n records
- if $\pi \in \mathbf{\Pi}$ is a transcript, the *posterior* $\mathcal{Q}_\pi$ is the conditional distribution:

$$\mathcal{Q}_\pi = (\mathcal{P}^n)|\mathrm{Interact}(M, \mathcal{A}; S) = \pi.$$

Roadmap

Sketch

1. The Bayesian resampling lemma implies that adaptive analysis is equivalent to non-adaptive analysis over posterior.
2. Relate analysis of dataset drawn from $\mathcal{P}^n$ to one drawn from the posterior through *posterior sensitivity*.
3. If an analysis has high in-sample accuracy and low posterior sensitivity, then it generalizes well.
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Bayesian resampling lemma

Lemma

Let $E \subset \mathcal{X}^n \times \mathbf{\Pi}$ be any event. Then:

$$\Pr_{S \sim \mathcal{P}^n, \Pi \sim I(S)} \left[(S, \Pi) \in E \right] = \Pr_{S \sim \mathcal{P}, \Pi \sim I(S), S' \sim \mathcal{Q}_\Pi} \left[(S', \Pi) \in E \right].$$

Bayesian resampling lemma

Proof sketch.

- Expand out the probability.

$$\Pr_{S\sim\mathcal{P},\Pi\sim I(S),S'\sim\mathcal{Q}_\Pi}\left[(S',\Pi)\in E\right]$$

$$=\sum_{\pi}\sum_{x'}\Pr[\Pi=\pi]\Pr_{S'\sim\mathcal{Q}_\pi}[S'=x']\mathbf{1}[(x',\pi)\in E].$$

- Apply Bayes rule.
- Collapse terms.

□

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Definitions

1. (ϵ, δ)-posterior sensitivity
2. (α, β)-sample accuracy and (α, β)-distributional accuracy
3. (ϵ, δ)-differential privacy

Definitions

Definition

An interaction $\mathrm{Interact}(M, \mathcal{A}; \cdot)$ *is* (ϵ, δ)-**posterior sensitive** *if for every data distribution* $\mathcal{P}$*:*

$$\Pr_{S \sim \mathcal{P}^n, \Pi \sim I(S)} \left[\max_j |q_j(\mathcal{P}^n) - q_j(\mathcal{Q}_\Pi)| \geq \epsilon \right] \leq \delta.$$

Definitions

Definition

*A statistical estimator M satisfies (α, β)-**sample accuracy** if for every data analyst $\mathcal{A}$ and every data distribution $\mathcal{P}$,*

$$\Pr_{S \sim \mathcal{P}^n, \Pi \sim I(S)} \left[\max_j |q_j(S) - a_j| \geq \alpha \right] \leq \beta.$$

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Definitions

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Two datasets $S, S' \in \mathcal{X}^n$ are **neighbors** *if they differ in at most one coordinate.*

Definition

*An interaction $\mathrm{Interact}(M, \mathcal{A}; S)$ satisfies (ϵ, δ)-***differential privacy*** *if for all data analysts $\mathcal{A}$, pairs of neighboring datasets $S, S' \in \mathcal{X}^n$, and for all events $E \subset \mathbf{\Pi}$,*

$$\Pr\left[I(S) \in E\right] \leq e^{\epsilon} \cdot \Pr\left[I(S') \in E\right] + \delta.$$

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General transfer theorem

Theorem (General transfer theorem)

Let $\mathrm{Interact}(M, \mathcal{A}; \cdot)$ *be* (α, β)*-sample accurate and* (ϵ, δ)*-posterior sensitive. Then, it is* (α', β')*-distributionally accurate, where* $\alpha' = \alpha + c + \epsilon$ *and* $\beta' = \frac{\beta}{c} + \delta$ *and* $c > 0$.

Accuracy over samples to accuracy over posterior

Lemma

Let M be (α, β)-sample accurate. Then for all $c > 0$,

$$\Pr_{S \sim \mathcal{P}^n, \Pi \sim I(S)} \left[\max_j |a_j - q_j(\mathcal{Q}_\Pi)| > \alpha + c \right] \leq \frac{\beta}{c}.$$

Accuracy over samples to accuracy over posterior

$$\Pr_{S\sim\mathcal{P}^n,\Pi\sim I(S)}\left[\max_j |q_j(S)-a_j| \geq \alpha\right] \leq \beta$$

$$\Downarrow$$

$$\Pr_{S\sim\mathcal{P}^n,\Pi\sim I(S)}\left[\max_j |a_j - q_j(\mathcal{Q}_\Pi)| > \alpha + c\right] \leq \frac{\beta}{c}.$$

Accuracy over samples to accuracy over posterior

Proof sketch. First, we obtain a one-sided tail bound:

$$\Pr_{S\sim\mathcal{P}^n,\Pi\sim I(S)}\left[\max_j a_j - q_j(\mathcal{Q}_\Pi) > \alpha + c\right]$$

$$\leq$$

$$\frac{1}{c}\mathop{\mathbb{E}}_{S\sim\mathcal{P}^n,\Pi\sim I(S)}\left[\Pr_{S'\sim\mathcal{Q}_\Pi}\left[\max_j a_j - q_j(S') > \alpha\right]\right],$$

almost directly from an application of **Markov's inequality**.

Accuracy over samples to accuracy over posterior

Proof sketch (cont). But we can rewrite this upper bound as the following probability:

$$\frac{1}{c} \Pr_{S \sim \mathcal{P}^n, \Pi \sim I(S), S' \sim \mathcal{Q}_\Pi} \left[\max_j a_j - q_j(S') > \alpha \right],$$

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which can be converted via the **Bayesian resampling lemma** into:

$$\frac{1}{c} \Pr_{S \sim \mathcal{P}^n, \Pi \sim I(S)} \left[\max_j a_j - q_j(S) > \alpha \right].$$

Accuracy over samples to accuracy over posterior

Proof sketch (cont). The same analysis holds for the other tail; combining them, we obtain the **two-sided tail bound**:

$$\Pr_{S\sim\mathcal{P}^n,\Pi\sim I(S)}\left[\left|\max_j a_j - q_j(\mathcal{Q}_\Pi)\right| > \alpha + c\right]$$

$$\leq$$

$$\frac{1}{c}\underbrace{\Pr_{S\sim\mathcal{P}^n,\Pi\sim I(S)}\left[\left|\max_j a_j - q_j(S)\right| > \alpha\right]}_{\leq\beta}.$$

□

Proof of general transfer theorem

Proof.

Lemma shows that if M is (α, β)-sample accurate, then

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If $\mathrm{Interact}(M, \mathcal{A}; \cdot)$ is (ϵ, δ)-posterior sensitive, then

$$\Pr_{S \sim \mathcal{P}^n, \Pi \sim I(S)} \left[\max_j |q_j(\mathcal{P}^n) - q_j(\mathcal{Q}_\Pi)| \geq \epsilon \right] \leq \delta.$$

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By the **triangle inequality**,

$$\Pr_{S \sim \mathcal{P}^n, \Pi \sim I(S)} \left[\max_j |q_j(\mathcal{P}^n) - a_j)| > \alpha + c + \epsilon \right] < \frac{\beta}{c} + \delta.$$

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1. Bayesian resampling lemma.
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Differential privacy implies low posterior sensitivity

Let M be a statistical estimator for a family Q of linear queries.

Lemma

If M is (ϵ, δ)-differentially private, then for any data distribution $\mathcal{P}$ and any analyst $\mathcal{A}$, it is (ϵ', δ')-posterior sensitive, for all $c > 0$ and $\epsilon' = e^{\epsilon} - 1 + 2c$ and $\delta' = \delta / c$.

Differential privacy implies low posterior sensitivity

Notation. If $S \sim \mathcal{P}^n$, let S_i be uniformly random record from S.

Proof sketch. Proceed by contradiction. We will aim to define an event $E \subset \mathcal{X}^n \times \mathbf{\Pi}$ so that a high posterior sensitivity implies:

$$\Pr_{(S_i,\Pi)}[(S_i,\Pi) \in E] - \Pr_{S_i \otimes \Pi}[(S_i,\Pi) \in E]$$

is large, but differential privacy implies that it is small.

Differential privacy implies low posterior sensitivity

Proof sketch (cont). Notice that this difference can be split apart in two ways:[1]

$$\Pr_{(S_i,\Pi)}[(S_i,\Pi) \in E] - \Pr_{S_i \otimes \Pi}[(S_i,\Pi) \in E]$$

[1] On *ad hoc* notation: let $E(\pi)$ be the set $\{x \in \mathcal{X} : (x,\pi) \in E\}$, and similarly for $E(\pi)$. Note that $E \subset \mathcal{X} \times \mathbf{\Pi}$.

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$$\Pr_{(S_i,\Pi)}[(S_i,\Pi) \in E] - \Pr_{S_i \otimes \Pi}[(S_i,\Pi) \in E]$$
$$= \sum_{\pi \in \mathbf{\Pi}} \Pr[\Pi = \pi] \sum_{x \in E(\pi)} \left(\Pr\left[S_i = x | \Pi = \pi\right] - \Pr[S_i = x]\right)$$

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&\quad = \sum_{x\in\mathcal{X}} \Pr[S_i=x]\big(\Pr\left[\Pi\in E(x)|S_i=x\right] - \Pr[\Pi\in E(x)]\big).
\end{aligned}$$

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\end{aligned}$$

The first relates well to posterior sensitivity while the second to differential privacy.

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Differential privacy implies low posterior sensitivity

Proof sketch (cont). Suppose that M has high posterior sensitivity, where:

$$\Pr\left[\max_j |q_j(\mathcal{Q}_\Pi) - q_j(\mathcal{P}^n)| > \alpha\right] > \frac{\delta}{c}.$$

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$$\Pr\left[\max_j |q_j(\mathcal{Q}_\Pi) - q_j(\mathcal{P}^n)| > \alpha\right] > \frac{\delta}{c}.$$

One of the tails must have at least half of the probability mass. Without loss of generality, assume:

$$\Pr\left[\max_j q_j(\mathcal{Q}_\Pi) - q_j(\mathcal{P}^n) > \alpha\right] > \frac{\delta}{2c}.$$

Differential privacy implies low posterior sensitivity

Proof sketch (cont). Define $\mathbf{\Pi}_\alpha$ to be the event:

$$\mathbf{\Pi}_\alpha = \left\{ \pi \in \mathbf{\Pi} : \max_j q_j(\mathcal{Q}_\pi) - q_j(\mathcal{P}^n) > \alpha \right\}.$$

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Proof sketch (cont). Define $\mathbf{\Pi}_\alpha$ to be the event:

$$\mathbf{\Pi}_\alpha = \left\{ \pi \in \mathbf{\Pi} : \max_j q_j(\mathcal{Q}_\pi) - q_j(\mathcal{P}^n) > \alpha \right\}.$$

These are the transcripts $\pi = \{(q_i, a_i)\}_{i=1}^k$ where one of the queries q_j distinguish between $\mathcal{Q}_\pi$ and $\mathcal{P}^n$ well. The previous slide just states that high posterior sensitivity implies:

$$\Pr\left[\Pi \in \mathbf{\Pi}_\alpha\right] > \frac{\delta}{2c}.$$

Differential privacy implies low posterior sensitivity

Proof sketch (cont). Recall the first decomposition:

$$\Pr_{(S_i,\Pi)}[(S_i,\Pi) \in E] - \Pr_{S_i \otimes \Pi}[(S_i,\Pi) \in E]$$
$$= \sum_{\pi \in \mathbf{\Pi}} \Pr[\Pi = \pi] \sum_{x \in E(\pi)} \left(\Pr\left[S_i = x | \Pi = \pi\right] - \Pr[S_i = x]\right).$$

Differential privacy implies low posterior sensitivity

Proof sketch (cont). Recall the first decomposition:

$$\Pr_{(S_i,\Pi)}[(S_i,\Pi) \in E] - \Pr_{S_i \otimes \Pi}[(S_i,\Pi) \in E]$$
$$= \sum_{\pi \in \mathbf{\Pi}_\alpha} \Pr[\Pi = \pi] \sum_{x \in E(\pi)} \left(\Pr\left[S_i = x | \Pi = \pi\right] - \Pr[S_i = x]\right).$$

If $E \subset \mathcal{X} \times \mathbf{\Pi}$ contains only transcripts from $\mathbf{\Pi}_\alpha$,

Differential privacy implies low posterior sensitivity

Proof sketch (cont). Recall the first decomposition:

$$\Pr_{(S_i,\Pi)}[(S_i,\Pi) \in E] - \Pr_{S_i \otimes \Pi}[(S_i,\Pi) \in E]$$
$$= \sum_{\pi \in \mathbf{\Pi}_\alpha} \Pr[\Pi = \pi] \sum_{x \in E(\pi)} \left(\Pr\left[S_i = x | \Pi = \pi\right] - \Pr[S_i = x]\right).$$

If $E \subset \mathcal{X} \times \mathbf{\Pi}$ contains only transcripts from $\mathbf{\Pi}_\alpha$, and the **inner sum** is $o(1)$,

Differential privacy implies low posterior sensitivity

Proof sketch (cont). Recall the first decomposition:

$$\Pr_{(S_i,\Pi)}[(S_i,\Pi) \in E] - \Pr_{S_i \otimes \Pi}[(S_i,\Pi) \in E]$$
$$= \sum_{\pi \in \mathbf{\Pi}_\alpha} \Pr[\Pi = \pi] \sum_{x \in E(\pi)} \left(\Pr\left[S_i = x | \Pi = \pi\right] - \Pr[S_i = x]\right).$$

If $E \subset \mathcal{X} \times \mathbf{\Pi}$ contains only transcripts from $\mathbf{\Pi}_\alpha$, and the **inner sum** is $o(1)$, then posterior sensitivity gives a lower bound:

$$\Pr_{(S_i,\Pi)}[(S_i,\Pi) \in E] - \Pr_{S_i \otimes \Pi}[(S_i,\Pi) \in E]$$
$$> o(1) \cdot \Pr\left[\Pi \in \mathbf{\Pi}_\alpha\right].$$

Differential privacy implies low posterior sensitivity

Proof sketch (cont). Construct E so that:

- The projection of E to $\mathbf{\Pi}$ is $\mathbf{\Pi}_\alpha$
- The terms in the inner sum is nonnegative

$$E(\pi) = \{x \in \mathcal{X} : \Pr\left[S_i = x | \Pi = \pi\right] - \Pr[S_i = x] > 0\}$$

Explicitly, define E to be:

$$E = \bigcup_{\pi \in \mathbf{\Pi}_\alpha} E(\pi) \times \{\pi\}.$$

Differential privacy implies low posterior sensitivity

Proof sketch (cont). If E defined this way, then:

$$\Pr_{(S_i,\Pi)}\left[(S_i,\Pi)\in E\right] - \Pr_{S_i\otimes\Pi}\left[(S_i,\Pi)\in E\right]$$
$$> o(1)\cdot\Pr\left[\Pi\in\mathbf{\Pi}_\alpha\right]$$

Differential privacy implies low posterior sensitivity

Proof sketch (cont). If E defined this way, then:

$$\Pr_{(S_i,\Pi)}\left[(S_i,\Pi)\in E\right] - \Pr_{S_i\otimes\Pi}\left[(S_i,\Pi)\in E\right]$$
$$> \alpha\cdot\Pr\left[\Pi\in\mathbf{\Pi}_\alpha\right]$$

we obtain a good lower bound.

Differential privacy implies low posterior sensitivity

Proof sketch (cont). On the other hand, if M is differentially private, then the second decomposition:

$$\Pr_{(S_i,\Pi)}[(S_i,\Pi) \in E] - \Pr_{S_i \otimes \Pi}[(S_i,\Pi) \in E]$$
$$= \sum_{x \in \mathcal{X}} \Pr[S_i = x]\big(\Pr[\Pi \in E(x) | S_i = x] - \Pr[\Pi \in E(x)]\big).$$

Differential privacy implies low posterior sensitivity

Proof sketch (cont). On the other hand, if M is differentially private, then the second decomposition:

$$\Pr_{(S_i,\Pi)}[(S_i,\Pi) \in E] - \Pr_{S_i \otimes \Pi}[(S_i,\Pi) \in E]$$
$$\leq \sum_{x \in \mathcal{X}} \Pr[S_i = x]\left[(e^{\epsilon} - 1)\Pr\left[\Pi \in E(x)\right] + \delta\right]$$

Differential privacy implies low posterior sensitivity

Proof sketch (cont). This provides an upper bound:

$$\Pr_{(S_i,\Pi)}\left[(S_i,\Pi)\in E\right] - \Pr_{S_i\otimes\Pi}\left[(S_i,\Pi)\in E\right]$$
$$< \left((e^{\epsilon}-1)+2c\right)\cdot \Pr\left[\Pi\in \mathbf{\Pi}_{\alpha}\right].$$

Differential privacy implies low posterior sensitivity

Proof sketch (cont). This provides an upper bound:

$$\Pr_{(S_i,\Pi)}\left[(S_i,\Pi) \in E\right] - \Pr_{S_i \otimes \Pi}\left[(S_i,\Pi) \in E\right]$$
$$< \left((e^{\epsilon} - 1) + 2c\right) \cdot \Pr\left[\Pi \in \mathbf{\Pi}_{\alpha}\right].$$

We obtain a contradiction if $\alpha \geq (e^{\epsilon} - 1) + 2c$. Thus, a differentially private mechanism must also have low posterior sensitivity. □

Transfer theorem for differential privacy

Theorem

Suppose that M is (ϵ, δ)-differentially private and (α, β)-sample accurate for linear queries. Then for every analyst $\mathcal{A}$ and $c, d > 0$, it is also (α', β')-distributionally accurate.[2]

[2] $\alpha' = \alpha + (e^{\epsilon} - 1) + c + 2d$ and $\beta' = \frac{\beta}{c} + \frac{\delta}{d}$.

Application to the Gaussian mechanism

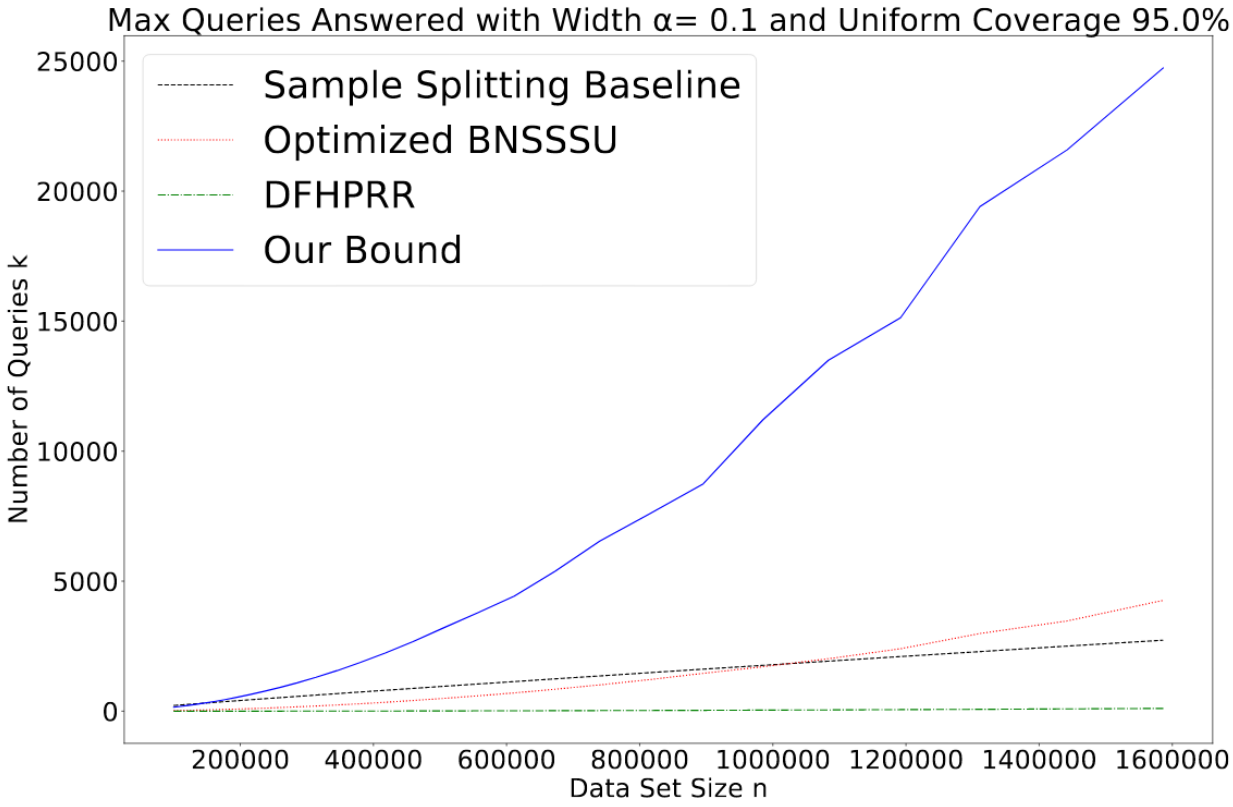

Figure 2: Comparison of lower bounds on the number of adaptive linear queries that can be answered using the Gaussian mechanism.

Extensions

- Recall in the proof that **accuracy over samples implies accuracy over posterior**, we made use of Markov's inequality to upper bound the error of analysis performed over the posterior. In the setting of $(\epsilon, 0)$-differential privacy, they obtain even better bounds by directly bounding the tail using a Chernoff-like concentration.
- In addition to linear queries, they also provide transfer theorems for **low sensitive** and **minimization** queries.

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Figures

1. Database icon made by Smashicons from www.flaticon.com.
2. Graph icon made by Freepik from www.flaticon.com.
3. Interaction LaTeXcode from https://tex.stackexchange.com/questions/211779/security-protocols-in-latex.

Appendix: accuracy over samples and posterior

Proof. Denote by $j^*(\pi) = \arg\max_j |a_j - q_j(\mathcal{Q}_\pi)|$.
Consider the one-sided tail probability:

$$\Pr_{S\sim\mathcal{P}^n,\Pi\sim I(S)} \left[a_{j^*(\Pi)} - q_{j^*(\Pi)}(\mathcal{Q}_\Pi) > \alpha + c\right].$$

Expanding out the expectation $q_{j^*(\Pi)}(\mathcal{Q}_\Pi)$, we get equality with:

$$\Pr_{S\sim\mathcal{P}^n,\Pi\sim I(S)} \left[\mathop{\mathbb{E}}_{S'\sim\mathcal{Q}_\Pi} \left[a_{j^*(\Pi)} - q_{j^*(\Pi)}(S') - \alpha\right] > c\right],$$

which is bounded above by:

$$\Pr_{S\sim\mathcal{P}^n,\Pi\sim I(S)} \left[\mathop{\mathbb{E}}_{S'\sim\mathcal{Q}_\Pi} \left[\left(a_{j^*(\Pi)} - q_{j^*(\Pi)}(S') - \alpha\right)_+\right] > c\right],$$

where $(x)_+ = \max\{0, x\}$.

Appendix: Bayesian resampling lemma

Proof.

Expanding out the probability:

$$\Pr_{S\sim\mathcal{P},\Pi\sim I(S),S'\sim\mathcal{Q}_\Pi}\left[(S',\Pi)\in E\right]$$

$$= \sum_{\pi}\sum_{x'}\Pr[\Pi=\pi]\Pr_{S'\sim\mathcal{Q}_\pi}[S'=x']\mathbf{1}[(x',\pi)\in E]$$

$$= \sum_{\pi}\sum_{x'}\Pr[\Pi=\pi]\Pr[S'=x'|\Pi=\pi]\mathbf{1}[(x',\pi)\in E]$$

$$= \sum_{\pi}\sum_{x'}\Pr[\Pi=\pi]\frac{\Pr[\Pi=\pi|S'=x']\cdot\Pr[S=x']}{\Pr[\Pi=\pi]}\mathbf{1}[(x',\pi)\in E]$$

$$= \Pr_{S\sim\mathcal{P}^n,\Pi\sim I(S)}\left[(S,\Pi)\in E\right],$$

where we made use of Bayes' rule. □

Appendix: accuracy over samples and posterior

Proof (cont). Markov's inequality yields the upper bound:

$$\frac{1}{c} \mathop{\mathbb{E}}_{S \sim \mathcal{P}^n, \Pi \sim I(S)} \left[\mathop{\mathbb{E}}_{S' \sim \mathcal{Q}_\Pi} \left[\left(a_{j^*(\Pi)} - q_{j^*(\Pi)}(S') - \alpha \right)_+ \right] \right].$$

Since $\left(a_{j^*(\Pi)} - q_{j^*(\Pi)}(S') - \alpha \right)_+ \leq 1$, a further upper bound:

$$\frac{1}{c} \mathop{\mathbb{E}}_{S \sim \mathcal{P}^n, \Pi \sim I(S)} \left[\Pr_{S' \sim \mathcal{Q}_\Pi} \left[a_{j^*(\Pi)} - q_{j^*(\Pi)}(S') - \alpha > 0 \right] \right],$$

which can be collapsed to:

$$\frac{1}{c} \Pr_{S \sim \mathcal{P}^n, \Pi \sim I(S), S' \sim \mathcal{Q}_\Pi} \left[a_{j^*(\Pi)} - q_{j^*(\Pi)}(S') > \alpha \right].$$

Appendix: accuracy over samples and posterior

Proof (cont). Applying the Bayesian resampling lemma, we see that the original one-sided tail probability is upper bounded:

$$\Pr_{S \sim \mathcal{P}^n, \Pi \sim I(S)} \left[a_{j^*(\Pi)} - q_{j^*(\Pi)}(\mathcal{Q}_\Pi) > \alpha + c \right]$$

$$\leq$$

$$\frac{1}{c} \Pr_{S \sim \mathcal{P}^n, \Pi \sim I(S)} \left[a_{j^*(\Pi)} - q_{j^*(\Pi)}(S) > \alpha \right].$$

This fills in the gaps in the above proof sketch. □